

supply output needs to charge a high-value capacitor, typically 20 μ F to 50 μ F in case of designators and rangefinders and as high as thousands of microfarads in high-power pulsed lasers, producing laser pulse energies of several kilojoules meant for electro-optical countermeasures and laser weapon applications.

Switched-mode concept is invariably used for the design of capacitor-charging power supply. Externally driven flyback converter is the preferred topology. Its design, however, is not as straightforward as it would be in the case of a resistive load. The reason is as follows.

In the case of capacitive load, energy storage and energy transfer mechanisms are relatively more complex. Each time an energy packet is stored in the primary of the transformer and subsequently transferred to the capacitor, the time needed to transfer the packet of energy depends upon the quantum of voltage it would impart to the capacitor.

As a result, for the same energy quantum, time required for transfer continuously reduces as the voltage builds up across the capacitor from zero to the final value due to diminishing voltage quantum. Therefore it is not advisable to use a fixed frequency or fixed off-time switching supply. The drive waveform needs to be a variable frequency one with the off-time periods governed by the charge status of the energy-storage capacitor.

It can be mathematically proved that the size of voltage packet varies from V when the capacitor is fully discharged to $[\sqrt{N}-\sqrt{(N-1)}] \times V$, where N is the number of packets required to charge the capacitor to the final value. If the off-time of the drive waveform could be varied or decreased to be more precise in accordance with the voltage build up across the capacitor, the power supply would operate at the highest possible conversion efficiency.

Fig. 2 shows the preferred block schematic arrangement of a capacitor-charging power supply configured around an externally-driven flyback converter.

The circuit is divided into two major parts, namely, the drive circuit and

the feedback circuit. The drive circuit comprises a cascaded arrangement of a voltage-controlled oscillator (VCO), a monoshot circuit and a drive circuit. Output of VCO feeds the trigger input of the monoshot. The pulse width of the monoshot is chosen to be equal to the desired on-time of the switching device. The frequency of monoshot output and hence the off-time of the waveform is governed by the frequency of the VCO output, which in turn depends upon the

voltage applied to its control input.

The drive circuit provides the required drive current and/or voltage depending upon the type of switching device used. The feedback circuit VCO output frequency is configured around a comparator and a subtractor. The comparator circuit is used to reset the monoshot circuit and therefore withdraw drive current from switching device as and when the output voltage reaches the desired output voltage.

The subtractor output controls the frequency of the VCO and therefore the off-time of the drive waveform to the switching device. The subtractor output in turn depends upon the voltage across the energy storage capacitor decreasing with increase in voltage.

Decrease in control voltage to VCO increases the output frequency, thereby reducing the off-time at the output of monoshot. What is important here is that the pattern of reduction of off-time is linearly related to the reduction of control voltage, which in turn is linearly related to increase in voltage across the capacitor. Thus off-time reduces in accordance with increase in the capacitor voltage. This design yields the most optimum results.

Compact capacitor-charging power supplies covering a range of input-output voltage and power output specifications are available from different manufacturers of laser electronics. One such unit intended for OEM applications is shown in Fig. 3.



Fig. 3: Capacitor-charging power supply module for OEM applications

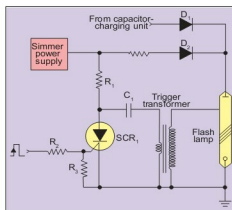


Fig. 4: Simmer power supply interfaced with a flash lamp

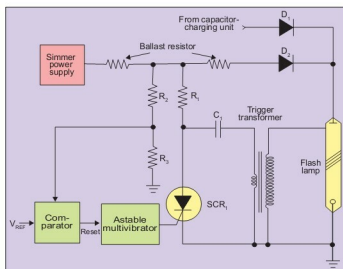


Fig. 5: Modified simmer interface circuit

Simmer power supply

The simmer power supply maintains a steady-state partial ionisation of the flash lamp during the time the lamp is not flashing by maintaining a low keep-alive current though it. Simmer power supply must be designed with due consideration to I-V characteristics of the flash lamp. The simmer power supply is a high-